

Advanced (Habanero)

Q1. Solve the equation $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve the equation $|2x| = 8$.

- (A) $x = 2$
- (B) $x = -2$
- (C) $x = 2$ or $x = -2$
- (D) $x = 4$
- (E) $x = -4$

Q3. Solve the equation $|x - 2| = 3$.

- (A) $x = 1$
- (B) $x = 5$
- (C) $x = 1$ or $x = 5$
- (D) $x = -1$
- (E) $x = -5$

Q4. Solve the equation $|x + 1| = 4$.

- (A) $x = 3$
- (B) $x = -5$
- (C) $x = 3$ or $x = -5$
- (D) $x = 1$
- (E) $x = -1$

Q5. Solve the equation $|3x - 2| = 10$.

- (A) $x = 2$
- (B) $x = 4$
- (C) $x = 2$ or $x = 4$
- (D) $x = -2$
- (E) $x = -4$

Q6. Solve the equation $|2x + 5| = 11$.

- (A) $x = 2$
- (B) $x = -8$
- (C) $x = 2$ or $x = -8$
- (D) $x = 1$
- (E) $x = -1$

Q7. Solve the equation $|x^2 - 4| = 8$.

- (A) $x = 2$
- (B) $x = -2$
- (C) $x = 2$ or $x = -2$
- (D) $x = 6$ or $x = -6$
- (E) $x = 4$ or $x = -4$

Q8. Solve the equation $|x - 3| = |2x - 5|$.

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 1$ or $x = 2$
- (D) $x = 4$
- (E) $x = 5$

Open Ended Questions (FRQ)

Q9. Solve the equation $|x + 2| = |3x - 1|$. Check for extraneous solutions.

Q10. Solve the equation $|2x + 1| = |x - 4|$. Check for extraneous solutions.

Chef's Tips

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- Remind students to consider both positive and negative solutions when solving absolute value equations.
- Emphasize the importance of checking for extraneous solutions.
- Encourage students to use algebraic methods to solve absolute value equations.